

★ OUTLINE + ROTATION VS REFLECTION ★

Reasoning | Ch06 Counting + Dice + Strategy | Topic 1 of 11

► Meaning in one line

Trace.

► Key points

Point	Example
trace.	follow perimeter.
rotation vs reflection.	turn keeps handedness; flip mirrors.

★ IMPORTANT FOR EXAM

Scanning + triangle-count

⚠ TRAP - read twice

Trace-miss + 'same shape' + turn/flip-mix

✓ **Try yourself:** Turn vs flip test? **Answer:** Write R: turn R stays R; flip makes mirror.

★ SCANNING + TRIANGLE COUNTING ★

Reasoning | Ch06 Counting + Dice + Strategy | Topic 2 of 11

► Meaning in one line

Sweep , formula.

► Key points

Point	Example
sweep.	systematic left-right-top-down.
formula.	count + formula-check.

★ IMPORTANT FOR EXAM

Square + rectangle grids

⚠ TRAP - read twice

Random-glance + 'counted' + formula-miss

✓ **Try yourself:** Scan + formula. Steps? **Answer:** Sweep + count + formula.

★ SQUARE + RECTANGLE COUNTING ★

Reasoning | Ch06 Counting + Dice + Strategy | Topic 3 of 11

► Meaning in one line

n^2 , sum-cubes.

► Key points

Point	Example
n^2 + sum-cubes.	3×3: $9+4+1 = 14$ squares.
L×W choices.	2×3 grid: 6×6? compute: $(2 \times 3/2) \dots$ use formula.

★ IMPORTANT FOR EXAM

Irregular + cube-net

⚠ TRAP - read twice

Eyeball + 'counted' + formula-miss

✓ **Try yourself:** Squares in 4×4? **Answer:** $16+9+4+1 = 30$.

★ IRREGULAR + CUBE NET ★

Reasoning | Ch06 Counting + Dice + Strategy | Topic 4 of 11

► Meaning in one line

Pieces.

► Key points

Point	Example
pieces.	split + count + sum.
fold + faces.	opposite never adjacent.

★ IMPORTANT FOR EXAM

Dice + standard-die

⚠ TRAP - read twice

Whole-guess + 'net obvious' + fold-miss

✓ **Try yourself:** Net + fold. Steps? **Answer:** Opposites + fold.

★ DICE VIEWS + STANDARD DIE ★

Reasoning | Ch06 Counting + Dice + Strategy | Topic 5 of 11

► Meaning in one line

Opposite , views.

► Key points

Point	Example
opposite from views.	common face + rotate.
clockwise + sum-7.	standard die check.

★ IMPORTANT FOR EXAM

Painted-cube + counts

⚠ TRAP - read twice

Adjacent-guess + 'opposite obvious' + rule-miss

✓ **Try yourself:** Standard die: 1 opposite? 2 opposite? **Answer:** 6; 5 (sum 7).

★ PAINTED CUBE + FACE COUNTS ★

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► Meaning in one line

Faces , formula.

► Key points

Point	Example
3/2/1/0-face.	corners = 3-face = 8.
$(n-2)^3$ zero-face.	n4: 8 zero-face.

★ IMPORTANT FOR EXAM

Partial + speed-rules

⚠ TRAP - read twice

All-painted + 'count visible' + formula-miss

✓ **Try yourself:** n5 painted all sides. 2-face count? **Answer:** Edges: $12 \times (n-2) = 36$.

★ PARTIAL PAINT + DIAGRAM-FIRST ★

Reasoning | Ch06 Counting + Dice + Strategy | Topic 7 of 11

► Meaning in one line

Cut.

► Key points

Point	Example
cut + faces.	paint-then-cut vs cut-then-paint.
diagram-first.	draw before solve.

★ IMPORTANT FOR EXAM

One-clue + strong-clue

⚠ TRAP - read twice

Imagine-only + 'obvious' + cut-miss + no-diagram

✓ **Try yourself:** Cut + diagram. Steps? **Answer:** Draw + cut + count.

★ ONE-CLUE + STRONG-CLUE PRIORITY ★

Reasoning | Ch06 Counting + Dice + Strategy | Topic 8 of 11

► Meaning in one line

Place , priority.

► Key points

Point	Example
one-clue + place.	fixed positions first.
strong/restrictive + priority.	most-constrained first.

★ IMPORTANT FOR EXAM

Reference + restriction-words

⚠ TRAP - read twice

Weak-first + 'any order' + stuck + no-priority = slow + wrong + fail! Definite-first.

✓ **Try yourself:** Puzzle + stuck + order? Fix? **Answer:** Definite-first + constrained.

★ REFERENCE + RESTRICTION WORDS ★

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► Meaning in one line

Anchor , start.

► Key points

Point	Example
anchor + start + build.	fix A, place rest.
restriction-words.	only/never/exactly constrain.

★ IMPORTANT FOR EXAM

Verification-pass + bailout

⚠ TRAP - read twice

Drift + 'probably' + restriction-miss + no-anchor

✓ **Try yourself:** Anchor + restrict. Steps? **Answer:** Fix + build.

★ VERIFICATION PASS ★

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► Meaning in one line

Recheck , clues , grid.

► Key points

Point	Example
Recheck + clues + grid.	-
Ex: every clue ticked + every box justified + final scan.	-

★ IMPORTANT FOR EXAM

Bailout + error-log

⚠ TRAP - read twice

Submit + 'done' + no-recheck + missed-clue

✓ **Try yourself:** Verify-pass + boxes. Steps? **Answer:** Clues + grid + scan.

★ BAILOUT + ERROR LOG ★

Reasoning | Ch06 Counting + Dice + Strategy | Topic 11 of 11

► Meaning in one line

30-45s , stuck , move.

► Key points

Point	Example
30-45s + stuck + move + mark.	-
categories + review + fix.	misread/calc/rush/guess.

★ IMPORTANT FOR EXAM

Full Reasoning shelf: 78pp + compendium + mocks + revision!

⚠ TRAP - read twice

Stuck-forever + 'one more minute' + no-log/review + repeat = time-death + fail! Clock + log.

✓ **Try yourself:** Stuck + sink + repeat-errors. Fix? **Answer:** Bail + log + review + fix.